

Optimal energy exchange of an industrial cogeneration in a day-ahead electricity market

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ABSTRACT

This paper addresses an optimal strategy for the daily energy exchange of a 22-MW combined-cycle cogeneration plant of an industrial factory operating in a liberalized electricity market. The optimization problem is formulated as a Mixed-Integer Linear Programming Problem (MILP) that maximizes the profit from energy exchange of the cogeneration, and is subject to the technical constraints and the industrial demand profile. The integer variables are associated with export or import of electricity whereas the real variables relate to the power output of gas and steam turbines, and to the electricity purchased from or sold to the market. The proposal is applied to a real cogeneration plant in Spain where the detailed cost function of the process is obtained. The problem is solved using a large-scale commercial package and the results are discussed and compared with different predefined scheduling strategies.

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1. Introduction

Cogeneration is extensively used as a distributed energy source to meet industrial thermal and electrical demand, one of the reasons being that cogeneration schemes offer high energy efficiency. In the last three decades governmental policies have strongly encouraged industries to install cogeneration as a way to obtain global energy savings and reductions in the environmental impact.

Under the scope of the ongoing restructuring process in electricity markets, cogeneration is now considered as a consumer/producer agent entitled to buy or sell electricity according to its needs, which in turn depend on the thermal and electric demands of the industrial process.

Some authors have proposed many ways to solve the optimal dispatch problem of cogeneration systems by minimizing all production costs (fuel, operation, maintenance, etc.) related to electric power and thermal capacity [1–7]. Pioneering contributions were made in the late 80s incorporating power trades between the cogeneration plants and electric distribution utility [8,9] inspired by the spot pricing theory condensed in Ref. [10]. The implementation of spot pricing theory derives from complex price schemes,

among other real-time prices (RTP) and time of use prices (TOU) [10,11]. An efficient strategy for energy exchange scheduling in a cogeneration plant is presented in Ref. [12] that considers real-time prices and consequently shifts steam operations.

To assess the impact of TOU prices in cogeneration sizing and operation several optimization procedures have been proposed in the literature and have been classified in Table 1.

In Ref. [13–16] Mixed-Integer Linear Programming (MILP) models have been proposed. Such optimization problems can be appropriately solved with large-scale commercial programming packages. Evolutionary Programming (EP) techniques as well as genetic algorithms (GA) were tested in Ref. [17–21] to tackle the nonlinearities of generation cost function. A Newton-based approach was proposed in Ref. [22] for optimal operation of back pressure cogeneration scheme under TOU rates.

In Ref. [23] a generalized formulation to determine optimal scheduling of a combined-cycle cogeneration unit is discussed, but does not properly consider an energy market structure. The economic benefit of energy exchange of cogenerators with the electric network under pool markets has not been extensively considered, while electricity markets are actually being worldwide operated on this basis.

Under day-ahead (DA) electricity markets the awareness of high or low electricity prices in the following day can encourage to sell or to buy electricity depending on the gas and steam turbines operation and the industrial demand in heat and electricity. The

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Table 1
Optimal cogeneration sizing and operation procedure

Type	Decision horizon	Solution technique
MILP	Short-term	Branch and Bound [13–16]
MINLP	Multi-time interval	Evolutionary Programming (EP) [17,18]
	Short-term	Genetic Algorithms (GA) [19,21]
	Short-term	GA and EP [20]
	Short-term	Newton based [22,23]

cogeneration plant can purchase power in several ways: directly from the DA market hour by hour, by a bilateral contract with a generator, or by a contract agreement with a retailer company (flat, two-part, TOU prices, etc.).

Operators of cogeneration systems face a complex problem when considering operating strategies for their facilities. The management of the cogeneration system requires constant evaluation of electrical and thermal load, plant performance variables and the relationship of these variables to energy prices. The operating strategy used must be dynamic and available on an uninterrupted basis to meet the management goal of maximizing the plant's economic return.

This work presents an optimal strategy for the energy exchange program of a cogeneration plant of an industrial factory in Spain entitled to sell power in a DA electricity market and to purchase electricity by a negotiated supply contract with a retailer. The optimization problem is written as a Mixed-Integer Linear Programming Problem maximizing the energy exchange profit of the cogeneration in the short term subject to the technical constraints as well as the industrial demand profile. The problem is solved using a large-scale commercial optimization package and the results are discussed and compared for different predefined scheduling strategies.

2. Combined-cycle cogeneration plant model

A combined-cycle cogeneration requires three basic pieces of equipment: gas turbine, waste heat recovery boiler and steam turbine. Figs. 1 and 3 show respectively the basic and detailed configurations. The cogeneration cost functions for gas turbine, steam turbine and waste heat recovery boiler are presented in the following.

2.1. Gas and steam turbines cost model

In a combined-cycle cogeneration scheme, the gas turbine cost function [23] is given by the following expression:

$$C_{GTh} = f \cdot M_{GTh} + C_{GTM} \quad (1)$$

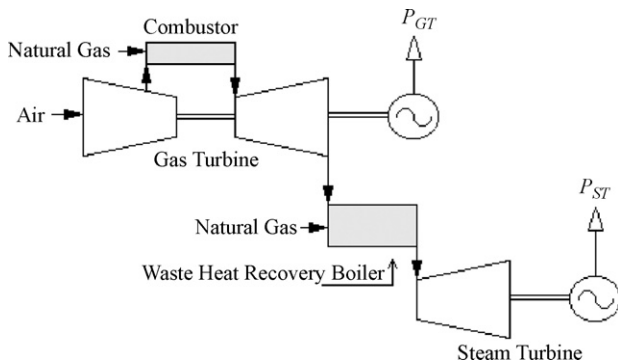


Fig. 1. Combined-cycle cogeneration basic configuration.

The mass flow rate of the fuel input M_{GTh} as a function of gas turbine power output P_{GTh} (in MW) is given by the following quadratic function:

$$M_{GTh} = aP_{GTh}^2 + bP_{GTh} + c \quad (2)$$

where a , b and c are the coefficients of the equipment provided by the gas turbine manufacturer.

The steam turbine cost per hour is only given by the maintenance cost $C_{STh} = C_{STM}$.

2.2. Waste heat recovery boiler model (WHRB)

The operating cost of the waste heat recovery boiler is related to the exhaust of gas turbines and the additional natural gas if required [23].

$$C_{Wh} = \left(M_{4h} \frac{H_4 - H_{10}}{\eta_W} - MF_h \cdot C_p \cdot \Delta T \right) \frac{f}{C_{vf}\rho} + C_{WM} \quad (3)$$

The exhaust gas flow rate MF_h as a function of gas turbine power output P_{GTh} at hour h is given by

$$MF_h = \phi + \gamma P_{GTh} \quad (4)$$

where ϕ (kg/h) and γ (kg/MWh) are specified values.

The output mass flow rate of steam M_{4h} (kg/h) as function of steam power output P_{STh} at hour h is given by:

$$M_{4h} = \sigma P_{STh} \quad (5)$$

where σ (kg/MWh) is a coefficient of the steam turbine.

3. Energy exchange framework

The energy exchange between distributed generators and electricity networks depends on the economic context of the liberalized market and on its regulatory rules. The main characteristics of the Spanish electricity market are presented below.

3.1. Energy imported from the electric network

The cost of imported energy for the industrial factory is given by a supply contract previously negotiated with a retailer that includes the traded energy price plus energy and capacity network access tariffs. These access tariffs are regulated and have two components, one depends on the energy actually supplied in a given hour and another on the power contracted. The network access tariffs are distributed in six periods and applied throughout the year taking into account the month and the type of day [24], as shown in Fig. 2.

The cost of imported electricity P_{IMPh} from the grid at given hour is written as:

$$C_{IMPh} = (C_{eh} + W_{eh}) \cdot P_{IMPh} + W_{Ch} + C_{EM} \quad (6)$$

3.2. Energy exported to market

In Spain, cogeneration is regulated under a special regime where all the energy injected to the electric network can be sold to the market [25]. The exported energy is paid at the hourly marginal price ρ_h previously established in the day-ahead electricity market plus an additional fee or premium pr . Price setting for exported energy to the market is prior to the optimization process and regarded as known variables. In this case, the plant is not able to modify the energy prices at the wholesale market. The revenues obtained by energy sold at given hour P_{EXPh} to the market are:

$$R_{EXPh} = (\rho_h + pr) \cdot P_{EXPh} \quad (7)$$

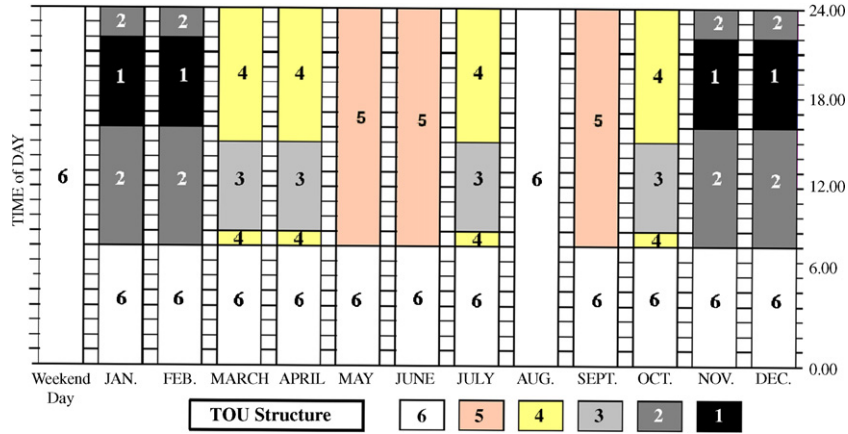


Fig. 2. Time of use structure of network access tariffs.

4. Formulation of the problem

The model is formulated as the maximization of profit from daily energy exchange defined as the difference among revenue by energy sold to the market and all operational costs. The variables of the optimization problem to be calculated are the operation condition (export or import), the amount of real power dispatched by the generating units and the amount of energy imported from the grid or exported to the market. Restrictions are related to the gas and steam turbine thermal constraints and the power balance for export and import condition. The problem formulated can be solved by commercial packages that are able to deal with Mixed-Integer Linear Programming models.

$$\begin{aligned} & \max (\text{Revenues} - \text{Costs}) \\ & \max \sum_{h=1}^{24} R_{\text{EXPh}} - \sum_{h=1}^{24} [C_{\text{GTh}}(P_{\text{GTh}}) + C_{\text{Sth}} + C_{\text{Wh}}(P_{\text{Sth}}, P_{\text{GTh}}) + C_{\text{IMPh}}] \end{aligned} \quad (8)$$

Subject to:

$$P_{\text{GTh}}^{\min} \leq P_{\text{GTh}} \leq P_{\text{GTh}}^{\max} \quad [\text{Gas Turbine Thermal Constraint}] \quad (9)$$

$$P_{\text{Sth}}^{\min} \leq P_{\text{Sth}} \leq P_{\text{Sth}}^{\max} \quad [\text{Steam Turbine Thermal Constraint}] \quad (10)$$

$$P_{\text{Dh}} = ((1 - \alpha)P_{\text{GTh}} + P_{\text{Sth}}) - P_{\text{EXPh}} + P_{\text{IMPh}} \quad [\text{Balance}] \quad (11)$$

$$P_{\text{EXPh}} \leq P_{\text{EXPh}}^{\max} \cdot I_{\text{EXPh}} \quad [\text{Maximum Exported Energy}] \quad (12)$$

$$P_{\text{IMPh}} \leq P_{\text{IMPh}}^{\max} \cdot I_{\text{IMPh}} \quad [\text{Maximum Imported Energy}] \quad (13)$$

$$I_{\text{EXPh}} + I_{\text{IMPh}} \leq 1 \quad [\text{Decision Variables Constraint}] \quad (14)$$

$$P_{\text{EXPh}}, P_{\text{IMPh}}, P_{\text{GTh}}, P_{\text{Sth}} \geq 0 \quad [\text{Non-negative Values}] \quad (15)$$

Gas and steam turbines must generate power within their thermal capacity limits, which are strongly dependent on out-door temperature. The gas turbine thermal capacity is defined at lower and higher operation limits by the turbine manufacturer (Eq. (9)). The steam turbine thermal capacity is defined at lower bounds by

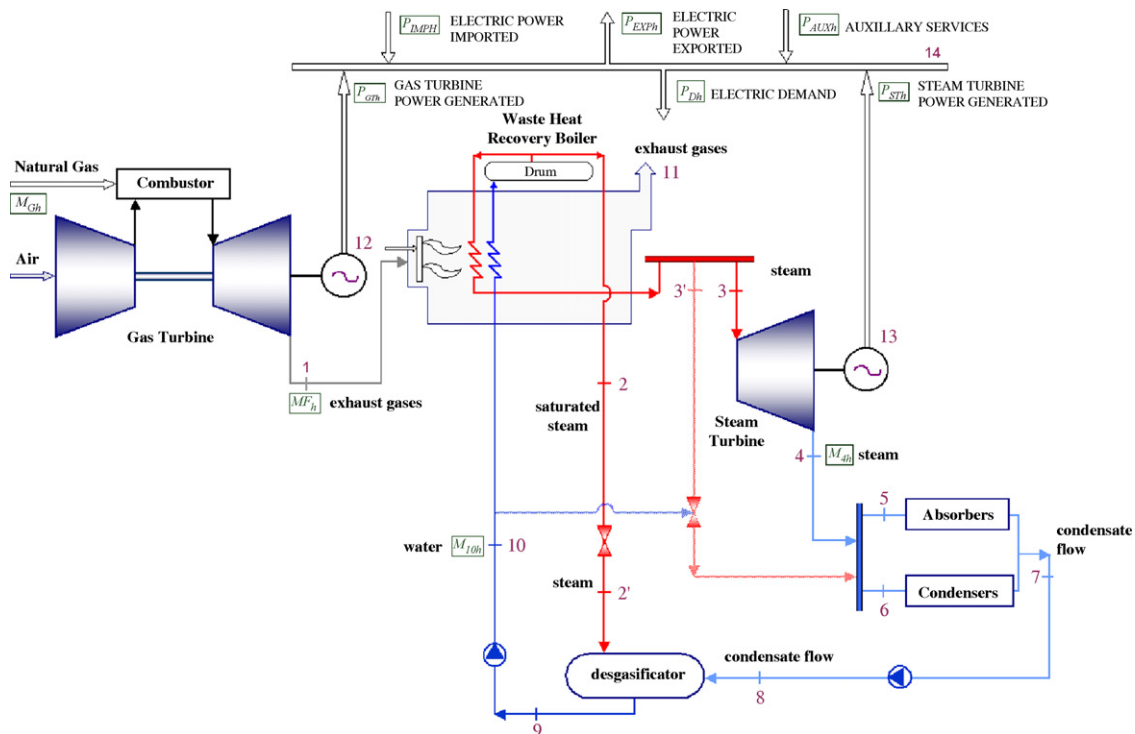


Fig. 3. Cogeneration plant scheme.

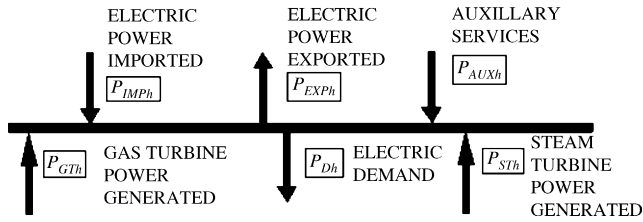


Fig. 4. Power balance.

the thermal demand of industrial productive process and at higher bounds by mechanical reasons (Eq. (10)).

The thermal demand of the industrial process must be satisfied by the cogeneration when it is in operation. The lower limit value of power generation of the steam turbine must always produce enough steam to supply the industrial thermal demand, introducing the output steam of the turbine in the absorbers for cooling water or in the condensers for hot water production. The correspondent diagram can be observed in Fig. 3, where the cogeneration plant scheme is shown. The thermal demand is coupled in the formulation by means of the equation of steam turbine power dependent on out-door temperature, which models the thermal requirements of the productive process. Data of real thermal demand and temperature along a year has been used to calculate the parameters of this equation (Eq. (19)). When the cogeneration plant is not in operation, additional chillers and boilers supply the thermal demand of the industrial process.

There are two feasible operation modes: importing power from the grid or exporting power to the grid, as shown in Fig. 4. This condition is identified using two binary variables I_{EXP} , I_{IMP} at each hour. The Eq. (14) imposes that only the export of electricity or only the import of electricity is possible at the same time in each hour.

The power of the auxiliary services of the gas turbine P_{AUXh} is given by the following expression:

$$P_{AUXh} = \alpha P_{GT_h} \quad (16)$$

Then, taking into account the Eq. (12) and Eq. (14), the export condition is verified when $I_{EXP} = 1$, and the Eq. (11) is as following:

$$P_{DH} = (1 - \alpha)P_{GT_h} + P_{ST_h} - P_{EXP_h} \quad (17)$$

and, taking into account the Eq. (13) and Eq. (14), the import condition is verified when $I_{IMP} = 1$, and the Eq. (11) is as following:

$$P_{DH} = (1 - \alpha)P_{GT_h} + P_{ST_h} + P_{IMP_h} \quad (18)$$

In order to establish a linear optimization problem, the non linear behavior of the gas turbine mass flow rate M_{GT} has been linearized using a partial derivative technique in four segments at the operating intervals as shown in Fig. 5.

5. Case study

The proposed methodology was applied to a real 22-MW combined-cycle cogeneration plant located in an industrial manufacturing factory in Spain. The detailed plant scheme is shown in Fig. 3. The gas turbine is rated 14.8 MW and the steam turbine is rated 8.1 MW. The waste heat recovery boiler can produce 19 ton/h of steam at 400 °C. The optimization model was applied in 48 scenarios (4 cases per month during 2006) taking into account whether the scenario was a working or a weekend day. Later, a general analysis is made on annual basis assuming that the optimization strategy is applied every day according to the Spanish calendar of network access tariffs. Finally, the results are compared with previous operation schemes.

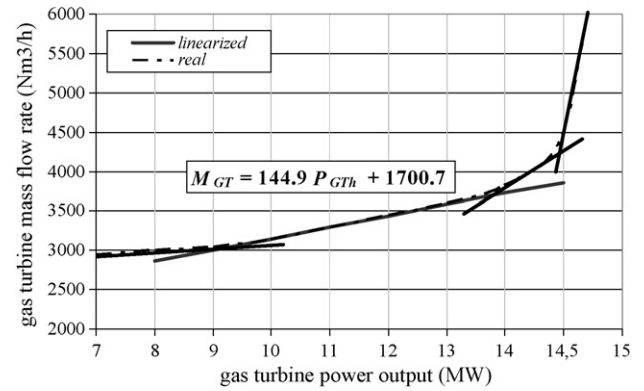


Fig. 5. Gas turbine mass flow rate as function of power output.

Commercial packages able to deal with Mixed-Integer Linear Programming models can implement the formulation of this problem [26]. In this case, the problem was solved using the CPLEX platform [27]. As a result, the binary variables I_{EXP} , I_{IMP} and the electric power dispatch variables P_{IMP_h} , P_{EXP_h} , P_{GT_h} , P_{ST_h} are obtained at each hour of the day. The daily revenue is obtained from Eq. (7) for 24 h of the day. Similarly, the cost of the imported energy and the equipment operational cost are obtained from the Eqs. (1), (2), (3), (6). The daily energy exchange profit is derived from the objective value (8).

5.1. Data

Table 2 presents the economic and technical data required to apply the model as obtained from the current plant operation.

The price of energy imported is known and the cogeneration plant has also an hourly selling price of exported electricity. The prices of the contract negotiated with a retailer for the energy import demand include all the components of the cost (the parameters of the Eq. (6) can be found in Table 2). On the other hand, the revenues by the exported energy are calculated from Eq. (7) as the hourly marginal price previously established in the day-ahead electricity market plus an additional fee or premium, according to the legal regulation of Spanish electricity market. The day-ahead prices of the electricity market used as hourly data ρ_h are shown in Figs. 6 and 7, and the numerical value of the additional fee ρ_r can be found in Table 2.

The thermal constraints of gas and steam turbines obtained from the operational sheets of the plant are specified as follows:

$$P_{ST_h}^{\min} = \begin{cases} 1.5 \text{ MW} & T \geq 34^\circ\text{C} \\ 5.190 - 0.1056 T \text{ MW} & -5^\circ\text{C} \leq T < 34^\circ\text{C} \end{cases} \quad (19)$$

Table 2
Economic and technical data

C_{GTM} (EUR)	79.80	ϕ (kg/h)	58204.8
C_{STM} (EUR)	6.96	γ (kg/MWh)	7740
W_{Ch} (EUR)	11.78	σ (kg/MWh)	6740
C_{EM} (EUR)	9.32	C_{VM} (EUR)	3.20
P_{EXPmax} (MW)	22.53	P_{IMPmax} (MW)	26.2
f (EUR/Nm ³)	0.14257	ΔT (°C)	330
pr (EUR/MWh)	11.97	C_{vr} (kJ/kg)	46095
$(Ce + We)_1$ (EUR/MWh)	59.20	ρ (kg/Nm ³)	0.8360
$(Ce + We)_2$ (EUR/MWh)	52.20	α	2.5%
$(Ce + We)_3$ (EUR/MWh)	51.00	b (Nm ³ /MWh)	144.9
$(Ce + We)_4$ (EUR/MWh)	48.70	c (Nm ³ /h)	1700.7
$(Ce + We)_5$ (EUR/MWh)	50.80	η_w	75%
$(Ce + We)_6$ (EUR/MWh)	31.50	C_p (kJ/kg °C)	1.2815
H_4 (kJ/kg) @120 °C, 1 BAR	2716.0	H_{10} (kJ/kg) @135 °C, 135 BAR	571.80



Fig. 6. DA electricity market and energy supply prices (January 3, 2006).

$$P_{STh}^{\max} = 8.1 \text{ MW} \quad -5^{\circ}\text{C} \leq T \leq 50^{\circ}\text{C} \quad (20)$$

$$P_{GTh}^{\min} = 13.4 \text{ MW} \quad -5^{\circ}\text{C} \leq T \leq 50^{\circ}\text{C} \quad (21)$$

$$P_{GTh}^{\max} = \begin{cases} 14.8 \text{ MW} & T < 8^{\circ}\text{C} \\ 15.805 - 0.1416T \text{ MW} & 8^{\circ}\text{C} \leq T < 17^{\circ}\text{C} \\ 13.4 \text{ MW} & T \geq 17^{\circ}\text{C} \end{cases} \quad (22)$$

To illustrate the results of the proposed methodology, two optimal power schedules of the 48 studied scenarios are presented:

- Scenario 1: January 3, Tuesday, 2006 (working day).
- Scenario 4: January 8, Sunday, 2006 (weekend day).

The data of hourly marginal prices corresponding to the DA market and the data of energy supply prices for both scenarios are presented in Figs. 6 and 7.

5.2. Working day: January 3 (Tuesday), 2006

In a working day, the maximum demand (26.2 MW) is reached between 6:00 and 13:00 h as shown in Fig. 8. Thermal constraints were fixed at average temperature equal to 8°C. The results reveal that the gas turbine is dispatched at maximum.

The steam turbine is dispatched close to the minimum allowable operating point due to the thermal constraints related to the outside temperature. No exported power is achieved during this working day. Imported power rises between 9:00 and 22:00 due to a slight increase in load demand.

5.3. Weekend day: January 8 (Sunday), 2006

In a weekend day load demand is low from 0.00 to 21.00 h (between 12 and 14 MW). However, the productive process begins



Fig. 7. DA electricity market and energy supply price (January 8, 2006).

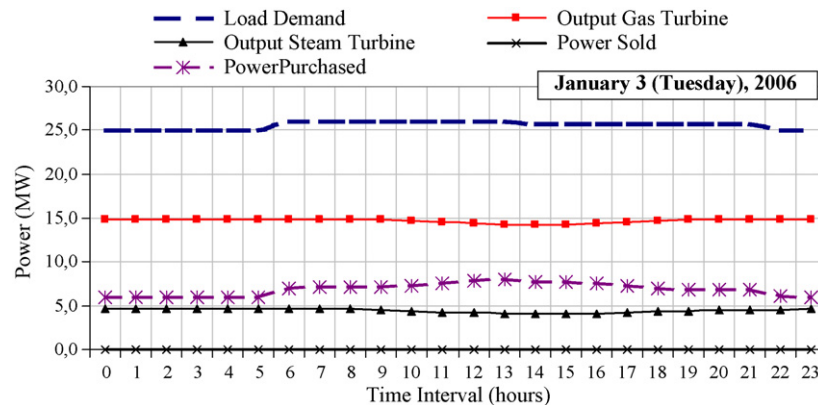


Fig. 8. Results of scenario 1.

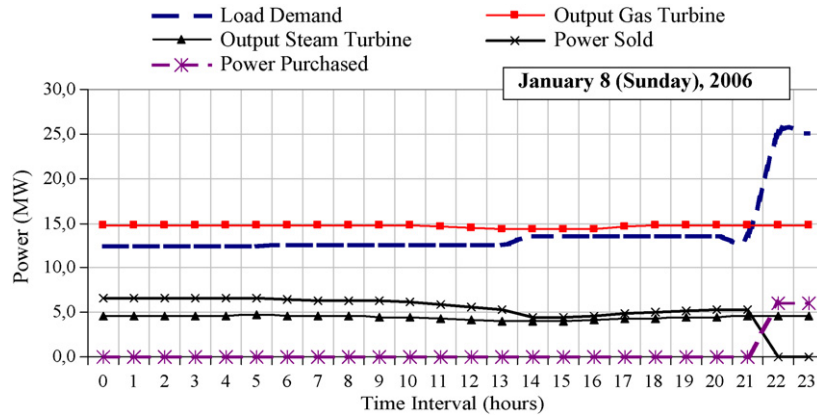


Fig. 9. Results of scenario 4.

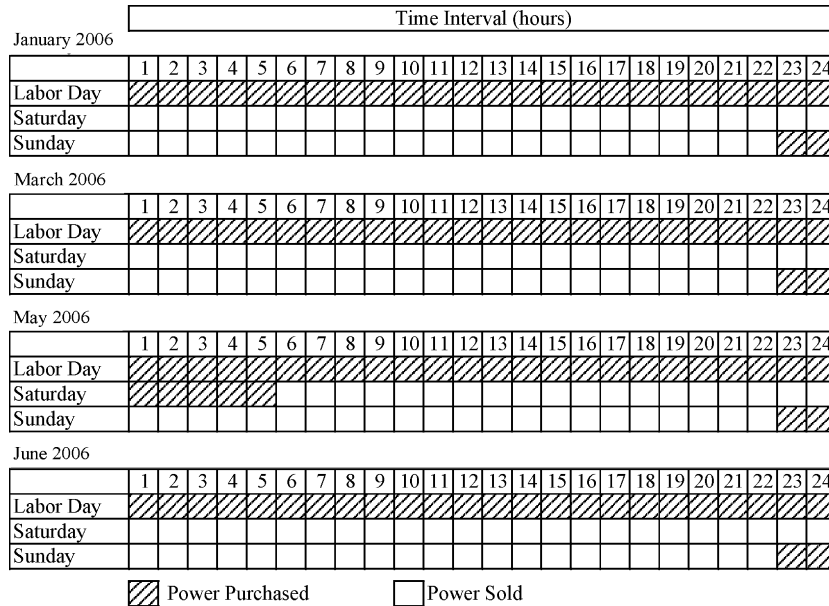


Fig. 10. Energy exchange scheduling diagram.

at 22:00 h and then load increases to 25 MW (Fig. 9). At this point, the results indicate that energy prices at DA market are very competitive with respect to the gas price. Gas and steam turbines are dispatched near to the minimum allowable operating point. It is observed in Fig. 9 an export condition between 0:00 and 22:00 h and an import condition between 22:00 and 24:00 h. It is interesting to point out that the power sold to the market does not have any correlation with the load shape.

5.4. Annual scenario results

The proposed methodology was applied on a daily basis during a whole year from Scenario 1 (January 1, 2006) to Scenario 48 (December 31, 2006). The optimal energy exchange can be traced by means of a scheduling diagram indicating when energy is imported or exported. Fig. 10 presents the results obtained for four months (January, March, May and June). The results distinguish between three types of day: working day, Saturday and Sunday.

Evidently, during working days no energy is exported to the grid: the industrial process consumes all the energy produced in the cogeneration unit. During the weekend there is a certain amount of energy generated that is exported to the grid. In some cases, on

Saturdays is observed that energy is imported in the 0:00–6:00 h interval and exported during the rest of day.

Fig. 11 shows a positive correlation between the imported energy and the market energy prices except for the month of August. In this month, the energy exported is maximum (2.9 GWh) and the energy imported is minimum (2.2 GWh) due to a decrease in the industrial activity and the high market prices, an average of 5.5 ctsEUR/kWh. In addition, it is important to point out that the

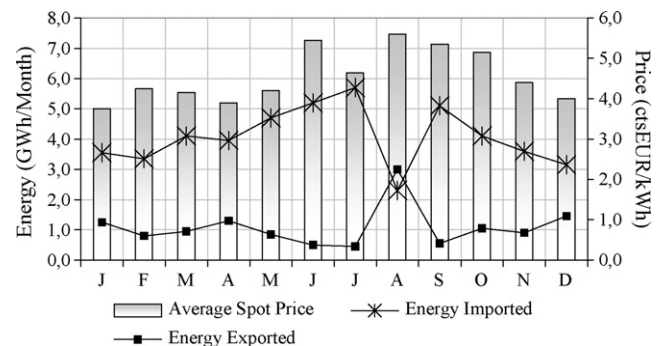


Fig. 11. Average spot price, energy exported and imported.

Table 3
Cogeneration turbine output operating points

Generation dispatch operation strategy		
Operation point	Type of day	
	Working	Weekend
1	Maximum	Maximum
2	Minimum	Minimum
3	Minimum	Maximum
4	Maximum	Minimum

maximum monthly amounts of imported energy occur in summer (5.2 GWh in June, 5.7 GWh in July and 5.1 GWh in September) when the average energy price is around 5.0 ctsEUR/kWh.

On the other hand, lower values of exported power are achieved during summer months (0.50 GWh in June, 0.45 GWh in July and 0.55 GWh in September). This behavior is due to the external temperature. The temperature in summer is higher than in winter limiting the gas turbine capacity. For instance, the gas turbine capacity in July and December is 13.4 and 14.8 MW, respectively, therefore the factory needs to import more energy in summer months due to generation capacity restrictions.

Thus, despite the energy price in summer being more attractive, the cogeneration plant is not able to export more energy due to these capacity constraints.

5.5. Comparative analysis

The economic dispatch results obtained for January 2006 by means of the optimal scheduling strategy is compared with four different operating points for the gas and steam turbines (see Table 3).

The operation strategy 1 considers that generators are dispatched at maximum allowable operation point in working days and weekend days. The operation strategy 2 involves minimum dispatching.

A mixed dispatch arrangement is considered in strategies 3 and 4. Table 4 shows the energy exchanged during January 2006, including a comparison with the hypothetical condition when cogeneration is turn off (importing all the energy from the grid).

Table 4
Energy exchange in January 2006

Operation point	Cogeneration plant					
	Input (GWh/month)			Output (GWh/month)		
	Production		Electricity purchased	Load demand	Auxillary services	Electricity sold
	Gas turbine	Steam turbine				
Optimum	10.09	2.82	4.27	16.17	0.25	0.75
1	9.65	2.82	4.64	16.17	0.24	0.69
2	10.16	5.83	2.02	16.17	0.25	1.59
3	9.79	3.62	4.60	16.17	0.24	1.59
4	10.39	5.03	1.71	16.17	0.26	0.69
Cogeneration off	0.00	0.00	16.17	16.17	0.00	0.00

Table 5
Energy exchange profit, revenues and costs in January 2006

Operation point	Cogeneration plant					
	Revenues (10 ³ EUR/month)		Costs (10 ³ EUR/month)		Profit	
	Electricity sold		Gas purchased	Electricity purchased	(10 ³ EUR/month)	Deviation (%)
Optimum	28.77		460.99	215.15	−647.37	–
1	26.19		457.18	230.26	−661.25	−2.10%
2	60.38		676.42	110.98	−727.03	−10.96%
3	60.38		515.64	228.96	−684.23	−5.39%
4	26.19		621.05	94.31	−689.17	−6.06%
Cogeneration off	0.00		0.00	725.79	−725.79	−10.80%

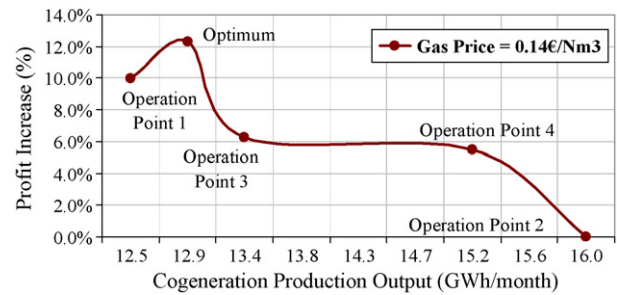


Fig. 12. Power profit as function of monthly cogeneration production output.

In the optimum strategy the gas turbine produces 10.09 GWh and the steam turbine produces 2.82 GWh; 4.27 GWh are imported from the grid and 750 MWh are sold to the market. Table 5 shows that the optimal energy dispatch produces a global electric power supply expenditure of 647.3 thousand EUR/month (intended as the difference between the costs of purchased gas and electricity and the revenues obtained for the electricity sold to the market). The optimal strategy solution is better than the other strategies as shown in Table 5 and Fig. 12. Note that minimum cost due to imported power is achieved under operation strategy 4 (94.31 thousand EUR/month) and maximum revenues are obtained under operation strategies 2 and 3 (60.38 thousand EUR/month). However, the best solution requires to spend 215.15 thousand EUR/month in electricity and 460.99 thousand EUR/month in gas, corresponding to the power dispatch provided by the optimization procedure.

It must be emphasized that major savings in energy expense are obtained when the optimal operation is compared with the operation strategy 2 (10.96%). Similar results are achieved when the cogeneration unit is not working or is out for maintenance (10.80%). These deviation percentages have been calculated for each operation strategy with respect to the best one (Table 5).

In Fig. 12 the profit increase of the five plotted points is calculated with respect to the worst operation point (point 2). In this case when the gas price is equal to 0.14 EUR/Nm³, the optimum opera-

tion strategy yields a profit increase of 12% respect to the operation point 2, 6% respect to the points 3 and 4, and 2% respect to point 1.

Finally, the impact of the gas price in the solution was analyzed and the results were obtained for a gas price reduction of 25% and a gas price increment of 25%. The profit increases if gas prices are lower, which involves a reduction in the global energy expenditure required by the productive process. Then, a reduction of 25% on gas prices leads to an increase of the average optimal generation of 3% from 13.4 to 13.8 GWh/month.

However, there are not any differences in optimal turbines power output with an increment of 25% in the gas price, and there is only a little decrease in the profit due to the increase of the gas price for the same energy production. The gas price of the cogeneration plant (0.14 EUR/Nm³) was a high price, so an increase of this price leads to the same operation points, typical of high gas prices.

6. Conclusion

This paper proposes a short-term optimization scheduling for a daily energy exchange of a combined-cycle cogeneration plant of an industrial factory. The knowledge in advance of day-ahead electricity prices facilitates the definition of an optimal strategy based on 24 h scheduling. The optimization problem is set as Mixed-Integer Linear Programming Problem and solved using a large-scale commercial package. The application of the model into a real cogeneration plant, using a very detailed cost function, shows an energy cost reduction of 10% (approximately 80.000 EUR/month) with respect to other predefined scheduling strategies.

Acknowledgements

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Appendix A. List of symbols

Nomenclature

C_{eh}	cost of the energy in hour h (EUR/MWh)
C_{EM}	maintenance cost of substation in hour h (EUR)
C_{GTh}	gas turbine cost in hour h (EUR)
C_{GTM}	maintenance cost of gas turbine in hour h (EUR)
C_{IMP}	cost of imported electricity in hour h (EUR)
C_p	calorific value of the exhaust gases (kJ/kg °C)
C_{STh}	steam turbine cost in hour h (EUR)
C_{STM}	maintenance cost of steam turbine (EUR)
C_{vf}	calorific value of additional natural gas (kJ/kg)
C_{Wh}	cost of waste heat recovery boiler in hour h (EUR)
C_{WM}	hourly maintenance cost of WHRB (EUR)
f	unit cost of fuel (EUR/Nm ³)
$H_4 H_{10}$	enthalpy of steam output and water (kJ/kg)
I_{EXPh}	binary variable of exported energy
I_{IMP}	binary variable of imported energy
M_{4h}	output mass flow rate of steam in steam turbine (kg/h)
M_{GTh}	gas turbine mass flow rate (Nm ³)
MF_h	exhaust gas flow rate into WHRB in hour h (kg/h)
pr	premium or additional fee (EUR/MWh)
P_{AUXh}	power of auxiliary services of the gas turbine
P_{Dh}	electricity demand in hour h (MW)
P_{EXPh}	power exported in hour h (MW)
P_{EXPmax}	constant of maximum power export (MW)
P_{GTh}	gas turbine power output in hour h (MW)
p_{min}^{max}	gas turbine capacity limits in hour h (MW)
P_{IMP}	power imported in hour h (MW)

P_{IMPmax}	constant of maximum power import (MW)
P_{STh}	steam turbine power output in hour h (MW)
p_{min}^{max}	steam turbine capacity limits in hour h (MW)
R_{STh}	revenues by energy sold to market in hour h (EUR)
R_{EXPh}	revenues by energy sold to market in hour h (EUR)
ΔT	drop in the exhaust gas temperature inside WHRB (°C)
Wc_h	capacity network access tariff in hour h (EUR)
We_h	energy network access tariff in hour h (EUR/MWh)

Greek symbols

α	power demand of the auxiliary services of the gas turbine (% of gas turbine power output)
ϕ	constant of waste heat recovery boiler (kg/h)
γ	constant of waste heat recovery boiler (kg/MWh)
η_W	WHRB efficiency
ρ	natural gas density (kg/Nm ³)
ρ_h	energy marginal price in hour h (EUR/MWh)
σ	constant of steam turbine (kg/MWh)

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[Fwd: Your submission]

4 mensajes

José María Yusta Loyo <jmyusta@unizar.es>**4 de enero de 2008 10:38**

Para: "Paulo M. De Oliveira-De Jesus" <pdeoliv@fe.up.pt>

Hola Paulo...

Por fin aceptaron el artículo de cogeneración... sólo he eliminado la frase que olvidé quitar en la revisión anterior y ya lo he remitido a la revista.

Un abrazo, José Mari

----- Mensaje reenviado -----

From: carloalberto.nucci@unibo.it

To: <jmyusta@unizar.es>

Date: Fri, 21 Dec 2007 18:03:09 -0000

Subject: Your submission

Ms. Ref. No.: EPSR-D-07-00154R1

Title: Optimal Energy Exchange of an Industrial Cogeneration in a Day-Ahead Electricity Market
Electric Power Systems Research

Dear Dr. JOSE M YUSTA,

Please submit to us a revised version of your above referenced paper. The attached referee comments were made. Once your revisions are made in accordance with these comments, we will be pleased to publish your paper.

If you decide to revise the work, please submit a list of changes or a rebuttal against each point which is being raised when you submit the revised manuscript.

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Yours sincerely,

Prof. Carlo Alberto Nucci
Associate/Regional Editor
Electric Power Systems Research

Reviewers' comments:

Reviewer #1: 1) The paper is on a very interesting and up-to-date subject

2) On paragraph 6 (page 9), it is better not to say that the presented cases are only two for lack of space,

as already advised in the first revision and apparently accepted by authors.

Reviewer #2: The paper has been improved. Furthermore, the authors have properly addressed all concerns raised by this reviewer. The paper can be published in the present form.

Paulo M. De Oliveira-De Jesus <pdeoliveira@usb.ve>

4 de enero de 2008 11:03

Para: José María Yusta Loyo <jmyusta@unizar.es>

Coñooooooooo por fin!!!
enhorabuena Josemari
año nuevo y noticias buenas
valio la pena los esfuerzos en Bruxelas, Liege, Oporto, Zaragoza y pare de contar!
Un abrazo y felicidades
Paulo

P.D: Quiero q vengas pronto por aca
mandame todos los temas que estas manejando
estoy preparando cursos para el 2008

2008/1/3, José María Yusta Loyo <jmyusta@unizar.es>:

[El texto citado se encuentra oculto]

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